

Thaier Hayajneh

**Introduction** To ensure U.S. leadership in AI, we must establish strong foundational pillars, starting with early AI education, workforce development, and strategic research investments. AI is rapidly advancing, and the U.S. must prepare students early to develop the skills necessary to compete with other nations and lead in AI innovation. The following policy actions are critical to achieving this goal.

**Priority Policy Actions for AI Education & Workforce Development**

- 1. Early AI Education as a National Priority** Develop a standardized AI curriculum in K-12 education, ensuring students gain foundational AI, machine learning, and computational thinking skills. Expand AI literacy programs in elementary and middle schools, integrating AI concepts into math, science, and technology education. Launch federal AI teacher training programs, equipping educators with the necessary knowledge and resources to teach AI effectively. Encourage public-private partnerships to provide AI learning resources, mentorship programs, and hands-on AI labs in schools.
- 2. AI Workforce Readiness & Global Competitiveness** Implement AI apprenticeships and industry collaborations to transition students from academic learning to real-world AI applications. Establish federal AI upskilling initiatives, particularly cybersecurity, data science, and ethical AI governance. Create a national AI talent pipeline, ensuring a diverse, skilled workforce ready to compete in the global AI landscape.

**Priority Policy Actions for AI Research & Development**

- 3. Strategic AI Investments for U.S. Leadership** Increase funding for ethical AI research, including bias reduction, explainable AI (XAI), and adversarial AI defense mechanisms. Support AI research in cybersecurity applications, focusing on fraud detection, threat intelligence, and national security. Establish federal AI innovation hubs to accelerate AI development in key sectors such as healthcare, defense, and finance.
- 4. National AI Security & Infrastructure Protection** Strengthen AI-driven cybersecurity solutions, ensuring the protection of critical infrastructure and sensitive data. Implement a federal AI risk assessment framework, ensuring AI applications align with national security priorities. Enhance U.S. AI competitiveness through strategic partnerships with academia, government, and industry.

**Conclusion** To establish the U.S. as a global AI leader, we must invest in early AI education, workforce development, and cutting-edge research. By building the foundational pillars of AI literacy, security, and innovation, we can ensure the U.S. remains the leading nation of AI advancement.